

KAVEESHA NIRMANI

AI/ML and Big Data Engineer

Galle, Sri Lanka | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Final year Computer Engineering student with hands on experience building applied AI/ML and big data systems end to end, including reinforcement learning, agentic RAG pipelines and MCP based tool orchestration, NLP, and computer vision, alongside large scale data engineering with Kafka, Spark, and Airflow, and cloud native, containerized deployment. Completed a six month AI/ML industrial training placement at Sri Lanka Telecom PLC. Comfortable taking a project from research and model design through to MLOps pipelines, deployment, and CI/CD.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript/TypeScript, SQL
- **AI/ML:** PyTorch, Reinforcement Learning (DDQN, PPO, MAPPO), Transformers, RAG, Hugging Face, scikit-learn, MediaPipe, Gensim (Word2Vec)
- **LLM/Agentic Tooling:** Agentic tool calling, Model Context Protocol (MCP), Groq API, ChromaDB (vector DB), sentence transformers
- **Backend and Web:** FastAPI, Flask, Node.js/Express, Next.js, REST APIs
- **Big Data and Streaming:** Apache Kafka, Apache Spark (Structured Streaming), Apache Airflow, Avro, Hadoop/MapReduce
- **Cloud and DevOps:** Docker, Kubernetes, GitHub Actions (CI/CD), Azure Cosmos DB, AWS (Cloud Practitioner track), RabbitMQ
- **Data and Databases:** PostgreSQL, MongoDB, Redis
- **Tools:** Git, Vercel

BIG DATA ENGINEERING PROJECTS

Fleet Operations Data Pipeline (Lambda Architecture)

Kafka · Spark Structured Streaming · Airflow · PostgreSQL · FastAPI · Docker

- Engineered a Lambda architecture pipeline for a simulated ride hailing fleet, separating a real time speed layer (Spark Structured Streaming, 30s micro batches, 1 minute tumbling windows) from a daily batch layer (Airflow, full recompute, 3 retry exponential backoff)
- Implemented Spark Structured Streaming consumers that validate telemetry schema, route malformed messages to a Dead Letter Queue, and upsert per vehicle state and windowed zone utilization metrics to PostgreSQL
- Built a FastAPI dashboard with an observability layer (heartbeat based health monitoring, automated alerts for idle vehicles, unprofitable vehicles, and DLQ error rate), fully containerized with Docker Compose

Kafka Order Processing Pipeline (Avro + DLQ)

Kafka · Avro · Dead Letter Queue · Python

- Built a Kafka producer consumer system for order processing using Avro schema serialization for structured, evolvable message contracts
- Implemented running average aggregation, retry logic, and a Dead Letter Queue to capture and inspect failed message processing

FEATURED PROJECTS

Hospital Bed Allocation and Queue Management System (Final Year Project)

Reinforcement Learning · PyTorch · Multi Agent RL · Azure Cosmos DB

- Designed and implemented a Multi Agent PPO (MAPPO) model using a CTDE architecture (shared actor network, centralized 60 dim critic) for fairness aware bed allocation and patient queueing across 4 hospital wards, benchmarked against a real world dataset from Karapitiya Teaching Hospital
- Built and fine tuned five RL approaches (DDQN, PPO, MARL, MARL DDQN, MAPPO) against an FCFS baseline; ran systematic hyperparameter sweeps (4 notebooks) and multi seed validation (5 seeds) to confirm training stability
- Diagnosed and resolved a temporal data alignment bug affecting MAPPO utilization, and independently built and debugged the production MLOps retraining pipeline (weekly MAPPO retraining and evaluation), fixing schema and evaluation bugs verified end to end in CI
- Ran feature ablation testing on the trained MAPPO model to identify the most fairness critical state features, alongside stress testing under increasing patient load

AI/ML Research Assistant: Agentic RAG System

Retrieval Augmented Generation · Agentic Tool Calling · MCP · Docker · CI/CD

- Built an end to end agentic RAG system that answers questions on AI/ML research by retrieving and grounding responses in real arXiv papers, reducing hallucination compared with direct LLM prompting
- Implemented an agent loop where the LLM decides whether to search a ChromaDB vector store or answer directly, with a hard cap on search attempts to prevent infinite tool call loops
- Exposed the research tools through a Model Context Protocol (MCP) server, tested live with Claude Desktop, alongside a FastAPI backend and web chat UI
- Containerized the app with Docker and built a GitHub Actions CI/CD pipeline that runs unit tests and smoke tests the container on every push

NewsLens: Automated News Summarization

NLP · T5 Fine Tuning · Word2Vec · FastAPI

- Fine tuned a T5 small transformer model for abstractive news summarization on the BBC News dataset, achieving a validation ROUGE L of 0.449, a notable gain over zero shot prompting
- Added Word2Vec based semantic search and word analogy exploration over the article corpus, served through a FastAPI backend with a self contained demo UI

Ceylon Compass: AI Sri Lanka Travel Guide

LLM Prompt Engineering · Flask · Mermaid.js

- Built a full stack AI travel chatbot that answers Sri Lanka travel queries and generates itinerary diagrams on request, using two purpose tuned LLM calls (Groq) behind a Flask backend
- Designed keyword + regex intent detection to trigger diagram generation only when relevant, with a 3 attempt retry policy and graceful fallback so the text answer never depends on the diagram succeeding

Large Scale Retail Sales Analysis (Big Data / MapReduce)

Hadoop · MapReduce · Docker · Distributed Data Processing

- Built a Hadoop 3.3.6 Streaming MapReduce job to compute total revenue and transaction counts per city across 1,000,000 retail transaction records
- Implemented a memory efficient reducer using a key change flush pattern (constant memory use) suitable for large scale, distributed processing

Event Ticket Booking Platform (Cloud Computing)

Microservices · Kubernetes · Docker · RabbitMQ · CI/CD

- Co built a cloud native microservices platform (auth, events, booking, notification services) with JWT authentication, Redis caching, and PostgreSQL persistence
- Designed the service boundaries using C4 modeling (Context/Container/Component) and implemented event driven communication via RabbitMQ fanout exchanges for asynchronous cross service notifications
- Configured Kubernetes manifests with HPA for auto scaling between 2 and 10 replicas
- Set up a GitHub Actions CI/CD pipeline and Dockerized the full stack for local (Docker Compose) and cluster deployment

EXPERIENCE

AI/ML Industrial Trainee, Sri Lanka Telecom PLC, Digital Platform Division 6 months

- Built a Sinhala Automatic Speech Recognition system using Whisper
- Developed an AI based Feature Request Analysis tool and a Bug Analysis system
- Built a Fiber Wire Length Calculation system using image processing techniques

EDUCATION

B.Sc. (Hons) in Computer Engineering

- Relevant coursework: Advanced AI (NLP, transformers, GANs, MLOps), Computer Vision and Image Processing, Cloud Computing (Kubernetes, microservices, virtualization), Big Data Engineering, Machine Learning, Artificial Intelligence
- Self directed study: Software Architecture, C4 modeling (Context, Container, Component, Code diagrams) and event driven design patterns; applied directly in the Event Ticket Booking Platform project above

REFERENCES

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